

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Ship 10+ production TypeScript features end-to-end: real-time event pipelines (Pusher), multi-model AI integrations (Claude, GPT-4o, Gemini), Auth0 RBAC, MCP tooling; 394 commits, 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

VectorLift	Semantic Search & Ranking Engine	2026
- Built production-grade semantic search platform: Kafka-driven indexing pipeline into Elasticsearch and Qdrant; FastAPI search API with per-stage latency tracking across 6 pipeline configurations (BM25, bi-encoder, cross-encoder re-ranker). - Fine-tuned sentence-transformers and MiniLM re-ranker on MS MARCO; evaluated with NDCG@10, MRR@10, MAP and paired bootstrap significance testing.		
Inventory Reservation Service	Transactional REST API	2026
- Built inventory reservation REST API in Java 21 and Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent endpoints, RFC 9457 error responses. - Verified concurrency correctness with Testcontainers: two concurrent reservations against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security, GitHub Actions CI.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js: MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; SSE live updates. - Validated with Testcontainers; benchmarked hot paths with k6; OpenTelemetry + Prometheus; GitHub Actions CI.		

Languages and Technologies

- Languages:** TypeScript, Python, Java, JavaScript, SQL, Bash
- Frontend & Full Stack:** React, Next.js, Node.js, NestJS, Spring Boot, REST APIs, SSE
- Data & Search:** PostgreSQL, MySQL, Redis, Elasticsearch, Qdrant, MongoDB, Kafka, Docker, Terraform
- Observability & Cloud:** OpenTelemetry, Prometheus, Grafana, k6, AWS, GCP, GitHub Actions

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog